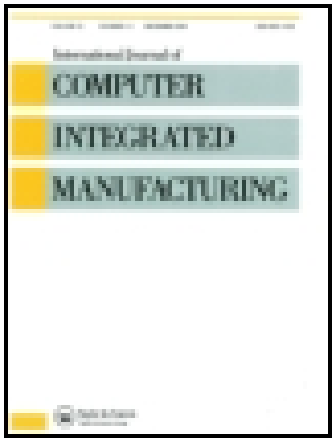
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High-speed machining (HSM) is an effective manufacturing process to produce parts. In HSM, it is required that the tool path should be smooth and the material removal rate should be constant. However, in geometry, it is nearly impossible to cover an arbitrary pocket with a single form of curves that satisfy both the above two requirements. In this paper, a compromise is made by subdividing the pocket into two kinds of regions: the HSM regions and the low-speed machining (LSM) regions. The HSM regions are selected to be the maximum inscribed circles (MICs) of the pocket. These MICs are calculated in an offset manner. Inside each HSM region, successive concentric circles are filled. The radii of the circles are controlled so that the material removal rate remains constant. The obtained concentric circles are then smoothly linked with pairs of arcs and used as the HSM tool path. For the rest LSM regions, conventional contour parallel tool paths are filled and low cutting speed is applied considering that there might be sharp angles on the pocket boundary. As the HSM regions could take up to 50% of the whole pocket and the cutting speed in HSM regions can be set very high, the average cutting speed for the whole pocket can be enhanced. Several pocket examples are used to verify the feasibility of the proposed HSM tool path generation method.

Keywords: computer-aided manufacturing; path planning; high-speed machining; region subdivision

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1. Introduction

Pocketing is a machining process which aims to clear out the totality of a predefined contour (Hatna, Grieve, and Broomhead 1998). It can be used as a stand-alone process in machining dies cavities, or as the layer-by-layer roughing process in machining free-form surfaces. Pocketing is often considered to be difficult for manual programming, especially for complex shapes, hence it relies highly on computer-aided manufacturing (CAM) systems. In CAM systems, the main task of pocketing is to fill inside a given planar shape with a series of tool paths. Depending on the data structure used, the shape can be represented by line segments, arcs, spline curves or a mixture of them. Generally, there are two basic filling strategies: direction-parallel (DP) and contour-parallel (CP).

DP tool path can be obtained by intersecting the contour with a series of parallel lines. For this kind of tool path, the inclination angle of the parallel lines and path element linking strategies (zigzag or one-way) have direct effects on the path efficiency (Park and Choi 2000). CP tool path can be generated by successively shrinking the contour. According to different shrinking methods used, CP path can be further divided into two types: contour-parallel offset (CPO) and spiral. Today, many techniques can be used to calculate the CPO tool path, for example, the pair-wise intersection detection (PWID) method (Choi and Park 1999), the Voronoi diagram method (Held, Lukács, and Andor 1994) and the straight skeleton method (Felkel and Obdrzalek 1998). In our previous research (Lin, Fu, He, et al. 2013), a linear offset algorithm based on profile updating rules is also proposed. On the Internet, sophisticated software packages on planar curve offset are freely available, such as Computational Geometry Algorithms Library (www.cgal.org) and Clipper (www.angusj.com/delphi/clipper.php). As for the comparison, the CPO tool path is generally considered to be more preferred to the DP tool path as it gives a better surface quality, less cutting time and less amount of slotting (Hatna, Grieve, and Broomhead 1998).

With the advances in machine tools and cutting tool technology, end milling at high rotation speeds, 'high-speed machining' (HSM), becomes a cost-effective manufacturing process to produce parts with high precision and surface quality (Fallböhmer et al. 2000). The definition of HSM changes with the material being machined. For example, a cutting speed of 500 m/min is considered HSM for cutting alloy steel whereas this speed is considered conventional in cutting aluminium. Significant advantages of HSM are high material removal rates, good surface finish, low cutting forces and minimal workpiece distortion (Toh 2006). However, the shortcomings of HSM include excessive tool wear, need for expensive machine tools with advanced spindles and controllers, and need for advanced cutting tool materials and

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coatings. Besides, HSM tools are equipped with very powerful feed drive systems, which can reach a high programmed feed rate at the expense of imposing significant dynamic constraints on the machine tool structure (Pateloup et al. 2006; Schmitz and Donalson 2000).

HSM imposes new challenges as well as difficulties for CAM tool path-planning task, especially for pocketing, since pocketing needs to remove most of the spare materials from the stock. In the context of HSM, in order to reduce tool chatter vibrations and tool wears, constant material removal rate should be maintained. Additionally, to meet the kinematic constraints and accommodate high feed rates, the tool path should be as smooth as possible, i.e. at least G^1-continuous. In an investigation on cutter path strategies in high-speed rough milling of hardened steel, Toh (2006) concluded that where long tool life, short machining times and high volumetric removal rates were concerned, one-way DP tool path was the most favourable one. However, in his investigation, a simple rectangular pocket is used throughout, and the CPO path he used is not optimised: the corners are not smoothed. To date, some corner rounding techniques for CPO tool path have been developed. With these techniques, Toh's conclusions might be different. Beside the above techniques, some other smooth tool path generation methods have also been proposed. In the following, the work related to smooth tool path generation or optimisation methods for pocketing will be briefly introduced.

1.1. Related work

In successive CPO process of planar curves, sharp corners are inevitable. In order to make the traditional CPO tool path adequate for HSM, these sharp corners must be smoothed. Realising this, Choy and Chan (2003) introduced a corner-looping-based tool path for pocket milling. In their work, bow-like tool path segments are appended to the basic CPO tool path at the corner positions so as to smoothen the cutter motion, as shown in Figure 1(a). Similarly, Zhao et al. (2007) also appended but with biarc transition segments to CPO corners, producing second-order continuous tool path, as shown in Figure 2(b). Note that in the above two methods, the radial cutting depths at corners are variable. To deal with such defect, Pateloup, Duc, and Ray (2004) calculated the radius value at CPO corners according to the kinematic behaviour of the machine tool and the radial depth of cut variation, as shown in Figure 1(c). In addition to this, they used B-spline to achieve the C^2-continuous tool path. Wang, Jang, and Stori (2005) developed an approach to improve the pocketing tool path based on two quantifiable process metrics: instantaneous path curvature and instantaneous cutter engagement. With their approach, the original CPO path gradually evolves into a new smooth path curve, as shown in Figure 1(d). Be noted that one possible side effect with the corner smoothing methods is the appearance of uncut areas. This must be carefully attended to.

In addition to the corner rounding techniques based on the CPO path, several smooth tool path generation methods have also been developed for HSM. Yao (2006) attempted to fill a 2D region with modified Archimedean spiral curves centred on the medial axis since Archimedean spirals have no sharp turns, as shown in Figure 2(a). Yao and Joneja (2007) further developed the above approach by using the combination of Archimedean and clothoid spirals. However, in these approaches, adjacent spirals may overlap. Bieterman and Sandstrom (2003)

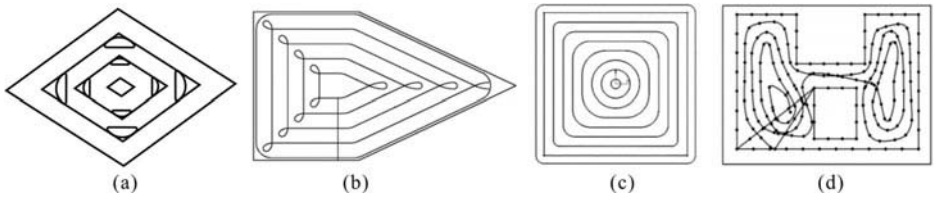


Figure 1. Different smoothing techniques for CPO tool path in the literature.

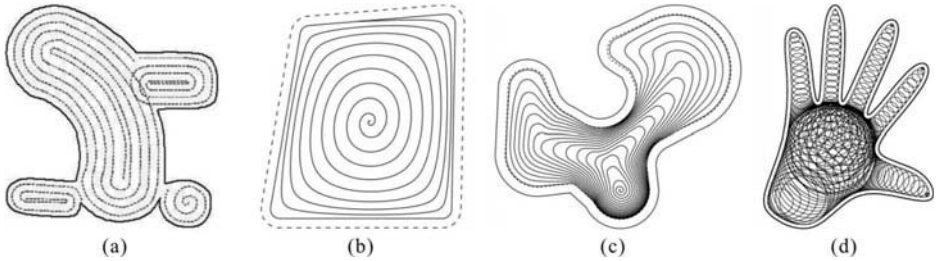


Figure 2. Different smooth tool path generation methods for HSM in the literature.

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proposed a curvilinear tool path generation method which uses the solution of an elliptic partial differential equation boundary value problem defined on a pocket region, as shown in Figure 2(b). This method morphs the pocket centre point to the pocket boundary and leads to substantial reductions of tool wear and machining time. Held and Spielberger (2009) pointed out that this method could only work for 'near convex' shapes; they introduced a new algorithm for generating G^1-continuous spiral tool path for HSM by interpolating growing discs placed on the medial axis of the pocket. This algorithm is capable of handling complex pockets and the generated tool path is guaranteed to be free of self-intersections, as shown in Figure 2(c). Later, Held and Spielberger (Held and Spielberger 2014) further developed the above algorithm by introducing a geometric heuristic for decomposing an arbitrary pocket with or without islands into simpler subpockets that are better suited for efficient spiral HSM. However, with their algorithms, constant material removal rate cannot be guaranteed. Be noted that in practical machining, the smoothness of above spiral tool paths can be achieved only if specific interpolators are installed in the numerical control unit (NCU) of the machine tools. Elber, Cohen, and Drake (2005) developed an interesting approach for HSM of pockets called medial axis transform towards HSM (MATHSM). In this approach, inscribed circles with variable radii are placed side by side along the medial axis of the pocket, as shown in Figure 2(d). These circles are directly used as HSM tool path after they are linked on the common tangent. However, MATHSM could generate about half non-cutting tool path and the material removal rate is non-constant, either. Similarly, Yamaji et al. (2003) used trochoid cycles to cover the pocket region.

1.2. *Achieved results*

Note that in the above smooth tool path generation methods, the aforementioned authors tried to fill the whole pocket with a single form of curves. Geometrically, for an arbitrary pocket, it seems impossible to find such a kind of curves that satisfy both the G^1-continuous and constant material removal rate conditions. The smoothness of the above tool paths are usually achieved by sacrificing the constant material removal rate condition. As pointed out by Coleman et al. (2008), it would be desirable for a CAM system to create tool paths for which the cutter engagement is maintained to be substantially constant, especially for HSM. In their work, they constrained the cutter engagement in a maximum given value.

In this work, the authors give up the idea of seeking such a kind of curves but try to fill the pocket with two kinds of curves. A new method of generating HSM tool path by region subdivision is presented. The input pocket is subdivided into two types of regions: HSM regions and low-speed machining (LSM) regions. In HSM regions, literally, it is demanded that the tool path should be at least G^1-continuous and material removal rate should be constant. In LSM regions, the above two conditions are less important since the machining speed is low, so conventional CPO curves can be filled.

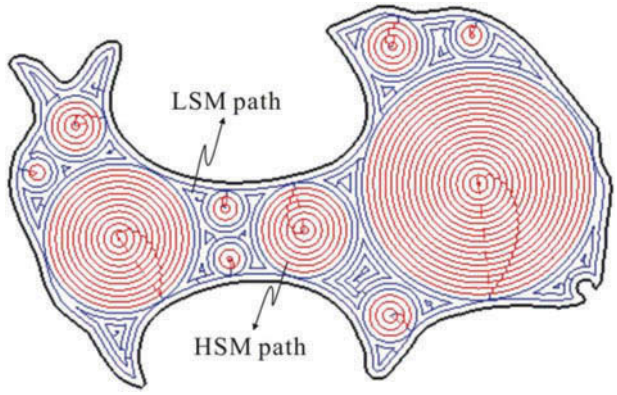


Figure 3. A demonstration of the proposed HSM tool path in a pocket.

For HSM regions, the region contours should be simple and smooth. So far, the authors have investigated several basic geometric shapes and find circles are the best, so circles are used for HSM regions. Inside each of these circular regions, a series of concentric circles are filled. The radius of each concentric circle is calculated so that the cutter engagement area from outer to inner circles remains constant. These concentric circles are linked smoothly with arcs and used as the HSM tool path. Another advantage of using circular path is that it is compatible with most of the NCUs. A demonstration of the proposed method is shown in Figure 3. From Figure 3, it can be seen that this method is especially suitable for fat pockets.

The rest of the paper is organised as follows. In Section 2, the details of subdividing a pocket into several circular regions are introduced. In these regions, smooth HSM tool paths with constant material removal rate are generated and linked in Section 3. Some examples are given in Section 4 and the last section concludes this paper.

2. Region subdivision

To increase the average machining speed, it is essential that HSM regions take most parts of a pocket. For each circular region, it is also essential that its area reaches maximum inside the current pocket. The problem can be considered as finding the maximum inscribed circle (MIC) inside an arbitrary 2D shape. Theoretically, the centre of MIC lies exactly on the medial axis of a shape. Although medial axis can be calculated with available tools (such as

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VRONI developed by Held (2001)), the problem is even with the obtained medial axis – it is still a trivial task to locate the MIC on it. In this section, a simple procedure for finding the MIC inside a shape with islands is developed as follows.

2.1. MIC calculation

Let's take the shape shown in Figure 4, for example. This shape contains an island. The direction of the outer contour is counterclockwise and the island clockwise. Under an offset at a certain distance R, the outer contour and the island are offset inwards and outwards, respectively. The two offset curves may intersect. In this case, they can be trimmed at the intersecting points and merged into several closed curves. The regions that are enclosed by the offset curves are called offset regions, as shown in Figure 4. Let's denote the total area of all the offset regions at R by $S_{or}(R)$. With the growth of R, the offset regions becomes smaller and smaller and will disappear one by one eventually. As the offset curves are actually the equidistance curves from the shape boundary, the following two facts exist: (1) the final disappearing site of the offset regions is actually the centre of the MIC of the shape; and (2) the critical offset distance R_c that makes the last offset region disappear is actually the radius of the MIC. With these facts, it is now easy to locate the MIC with a simple bisection method.

Before the bisection starts, it is necessary to determine of the lower and upper limits of R: R_{lower} and R_{upper}. At R_{lower}, the offset region exists; while at R_{upper}, the offset region disappears. Consider the fact that the diameter of the MIC must be shorter than the minimum side length (denoted by $L_{\min}$, see Figure 4) of the axis aligned rectangle of the shape, so initial R_{upper} can be set to $L_{\min}/2$. With initial R_{upper} obtained, the initial R_{lower} is set to be $R_{\text{upper}}/2$. The obtained R_{lower} is then used to offset the shape. If offset region exists, then start the bisection with current R_{lower} and R_{upper} values; otherwise further set $R_{\text{upper}} = R_{\text{lower}}/2$ and $R_{\text{lower}} = R_{\text{upper}}/2$ until the offset region for R_{lower} exists.

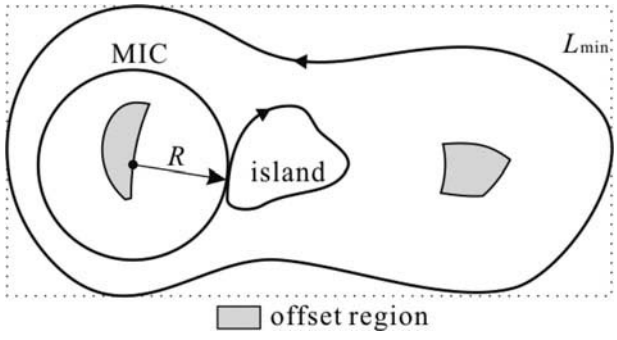


Figure 4. A shape with an island inside.

With a simple bisection, the approximate critical R_c can be found between $[R_{\text{lower}}, R_{\text{upper}}]$ in a few iterations. The stop condition of the bisection can be the satisfaction of the area $S_{or}(R_c)$ smaller than a prescribed area resolution S_ε. At offset distance R_c, there might be one or more last offset regions. The centre of the MIC can be randomly picked on a random offset curve.

Note that the above MIC calculation method is an approximate method. The precision of the result depends on the prescribed area resolution S_ε. For tool path planning tasks, S_ε can be defined according to the size of cutting tool used. The iterations in the bisection depend on both the shape size and S_ε. As the time complexity of the offset algorithm is $O(n\log n)$, in which n is the point size of the shape (including islands), the time complexity of the MIC calculation method is $O(m{\cdot}n\log n)$, in which m is the total offset times in and before the bisection. m is usually not very big, say less than 10 or 20, so the time complexity of the MIC calculation method is acceptable.

2.2. Region subdivision

With the above MIC calculation method for a 2D shape, it is now easy to carry on the region subdivision task. For the input pocket, it is used as the input of the MIC calculation process in Section 2.1; the outputs are the centre and radius of the MIC. Let's denote the radius of the obtained MIC as R_{c1}. R_{c1} is reduced to R_1 as follows:

$$R_1 = R_{c1} - 2\rho \tag{1}$$

where ρ is the radius of the cutter used. The concentric circle of the MIC with radius R_1 is defined as a disc. The MIC is embedded as an island into the input pocket. The number of islands inside the pocket increases by 1. The updated pocket is then reused as the input pocket.

The above process repeats until a stop condition is met. The output is a set of discs with known centre positions and radii inside the original pocket, as shown in Figure 5. The stop condition can be the satisfaction of R_i smaller than a predefined radius value R_ε, in which R_i is the radius of the generated disc in the ith iteration. The regions in the discs are used as the HSM regions, in which HSM is applied. The residual regions of the pocket are used as the LSM regions, in which conventional CPO tool path are filled.

For the LSM regions, the first offset distance from the region boundaries must be set to the cutter radius ρ. More specifically, as shown in Figure 5, the first circular curves outside the discs are used to clear out the material around the discs. For other offset distances, they are uniformly set to be ω:

$$\omega = k\rho \tag{2}$$

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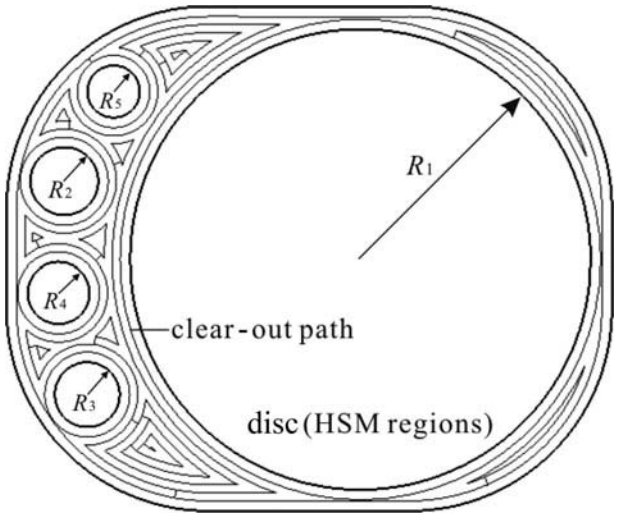


Figure 5. Discs inside a pocket.

where k is a coefficient. It is advised that k should be set equal to or smaller than 1 to prevent uncut phenomenon on CPO corners (Lin, Fu, Shen, et al., 2013). The generated CPO curves are linked according to the strategy proposed by Park, Chung, and Choi (2003). The strategy can be simply described as first constructing the tree structures of all the offset curves according to their inclusion relations and then linking the curves on the child node to the parent node. With their linking strategy, minimum tool retraction times can be achieved. As a matter of fact, the tool retraction times are equal to the splits occurred on the first offset curve.

For the HSM regions, concentric circles are filled and smoothly linked as the HSM tool paths. The details are explained in the following section.

3. HSM path generation

Held and Spielberger (2009) define the cutting width, δ, at a particular tool location as the directed Hausdorff distance from the previous cutting border to the current cutting border, as shown in Figure 6. With this definition, constant cutting width does not necessarily mean constant material removal rate, especially for the continuous cutting process of a disc, where the radius of the disc shrinks in several cuts, as shown in Figure 7. In HSM, a set of concentric circles, if constant material removal rate is to be achieved, the cutter engagement area should be controlled as constant during the machining process from outer to inner of the disc. This task can be described as follows: for the given disc with radius R and cutter with radius ρ, find the radius of the cutter path circle (denoted by r) that makes the cutter engagement area equal to a prescribed area S_p, as shown in Figure 6.

3.1. Path circle calculation

As shown in Figure 7, the cutter engagement area S_{ea} can be formulated as

$$S_{ea} = \alpha R^2 - \frac{1}{2}R^2 \sin(2\alpha) sign(\alpha) + \beta\rho^2 - \frac{1}{2}\rho^2 \sin(2\beta) sign(\beta) \quad (3)$$

where

$$\alpha = \arccos\left(\frac{R^2 + r^2 - \rho^2}{2Rr}\right),\ \beta = \arccos\left(\frac{\rho^2 + r^2 - R^2}{2Rr}\right)$$

and $sign(x)$ is function defined as follows:

$$sign(x) = \begin{cases} 1, & 0 < x \le \pi/2 \\ -1, & x > \pi/2 \end{cases}$$

With the given disc radius R and cutter radius ρ, S_{ea} can be simply represented as a function of r, namely $S_{ea}(r)$. From Figure 7, obviously, $S_{ea}(r)$ is a monotone increasing function and r ranges from $R - \rho$ to $R + \rho$. Let $S_{ea}(r)$ be equal to a prescribed cutter engagement area S_p, namely

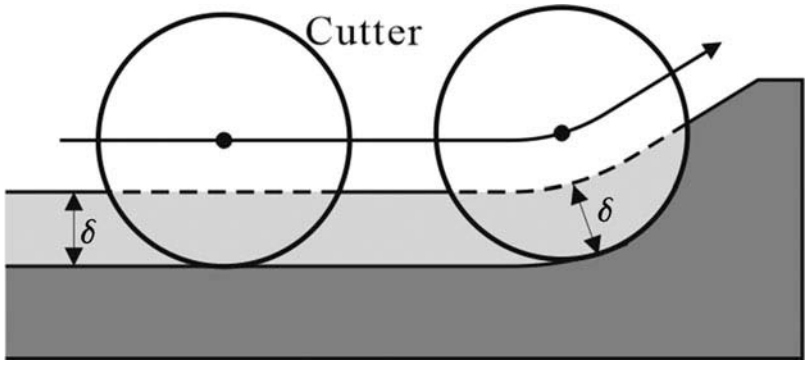


Figure 6. Cutting width definition in Held and Spielberger (2009).

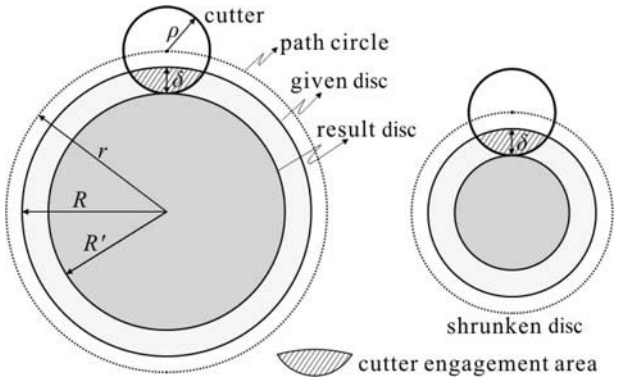


Figure 7. Cutter engagement area of different stages in machining a disc.

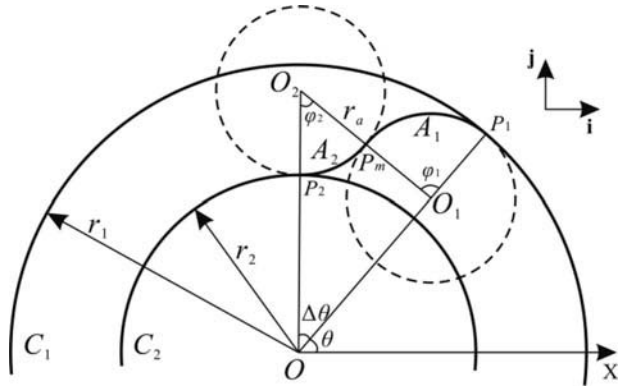


Figure 8. Linking two path circles with two arcs of same radii

$$S_{\mathrm{ea}}(r) = S_{\mathrm{p}} \tag{4}$$

In Equation (4), r can be easily solved with a numerical method, so the path circle can be determined.

When the cutter cuts the given disc along the path circle, as shown in Figure 7, a new disc is resulted. The resulting disc is used as the given disc and the next path circle that makes the cutter engagement area equal to S_{p} can be determined. This process repeats until the final disc region is smaller than the S_{p}. The output is a set of path circles. When the cutter moves along these circles, the material removal rate is constant. However, these circles need to be further linked before they can be used.

3.2. *Path linking*

Let's take two adjacent path circles (denoted by C_1 and C_2), for example, and denote their radii as r_1 and r_2 ($r_1 > r_2$), respectively. The problem here is to find the linking arcs that can smoothly link the two circles. Theoretically, there are many possible solutions since the number of the arcs used and the radius of each arc are not given. To make it easier, the authors choose the possible solution with two linking arcs of same radii (denoted by r_a). Suppose the value of r_a is given, then the problem becomes finding the centres and radians of the two arcs.

As shown in Figure 8, the link-out point on C_1 is denoted by P_1 and the link-in point on C_2 is denoted by P_2. Suppose the position of P_1 has been decided in the previous step. The angle formed by OP_1 and the X-axis of the planar coordinate system is denoted by θ. P_1 is the start point of the first arc (denoted by A_1) and P_2 is the end point of the second arc (denoted by A_2). Suppose the radians of A_1 and A_2 are φ_1 and φ_2, respectively, and the joining point of A_1 and A_2 is P_m. To make it smooth, the corresponding pairs of tangents on P_1, P_m and P_2 should share the same directions.

In Figure 8, O is the common centre of C_1 and C_2, O_1 and O_2 are the centres of arcs A_1 and A_2, respectively. As shown in Figure 8, in triangle OO_1O_2, the lengths of three sides are $r_1 - r_a$, $2r_{\mathrm{a}}$ and $r_2 + r_a$, which are all known, so values of $\Delta\theta$ and φ_2 can be calculated with arccosine functions. With $\Delta\theta$ and φ_2, $\varphi_1 = \Delta\theta + \varphi_2$. The position of O_1 and O_2 can be calculated as follows:

$$O_1 = O + (\boldsymbol{i}\cos\theta + \boldsymbol{j}\sin\theta)\cdot(r_1 - r_a) \tag{5}$$

$$O_2 = O + (\boldsymbol{i}\cos(\theta + \Delta\theta) + \boldsymbol{j}\sin(\theta + \Delta\theta))\cdot(r_2 + r_a) \tag{6}$$

in which $\boldsymbol{i}$ and $\boldsymbol{j}$ are the unit vectors of X- and Y-axis directions of the coordinate system, respectively.

Suppose the two path circles C_1 and C_2 are both in counterclockwise directions, then arc A_1 should be in counterclockwise direction and arc A_2 should be in clockwise direction. Now the only problem left is the decision of r_a. Obviously, r_a should be related to both r_1 and r_2. In this work, to make it simple, r_a is defined as

$$r_a = \frac{r_1 - r_2}{2} \tag{7}$$

With the above discussion, the two linking arcs can be completely determined. P_2 is then used as the linking point for the next pair of arcs. In this way, from a selected start point (denoted by P_s) on the outmost path circle, the circles are linked one by one until to the last. Then the cutter moves directly to the centre of the disc to ensure no material is left there. After that, the cutter rapidly moves back to P_s, as shown in Figure 9. It must be noted that when the cutter moves on the linking arcs, the material removal rate is non-constant.

The linked HSM path is further linked to the LSM CPO curves at a nearest point to P_s, as shown in Figure 9. In this way, all the HSM and LSM paths can be linked as one.

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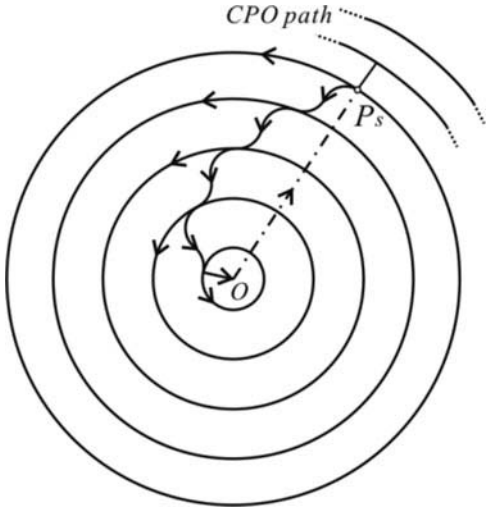


Figure 9. Linked circular path in the HSM region.

4. Examples

The proposed HSM tool path generation method in this work is implemented in C++ programming language. In the test program, the cutting speed of the LSM and HSM tool paths can be separately set. Generally, the cutting speed in HSM regions are much higher than that in LSM regions. In LSM regions, the offset distances are set according to the cutting width (see Equation (2) and Figure 6); while in HSM regions, the radii of the path circles are controlled by a given cutter engagement area S_p (see Equations (3) and (4)). Several arbitrary pockets are used to test the proposed method.

Figure 10 shows the first pocket example including an island inside. The size of the pocket is about 120 mm × 120 mm. For this test, an end mill with radius 2 mm is used to generate both the LSM and HSM tool paths. The k value as in Equation (2) is set to be 0.8, namely the cutting width in LSM regions is 1.6 mm. For the HSM regions, the cutter engagement area S_p is set to be 4.68 mm^2 and remains a constant. The generated tool path is shown in Figure 10. For this example, 11 circular HSM regions are extracted from the original pocket. The maximum HSM region has a radius of 28.3 mm while minimum HSM region has a radius of 2.04 mm. The total area for all the HSM regions is about 4492 mm^2, which takes about 45% of the whole pocket area. This percentage can still be increased if a smaller area resolution S_ε is set (see Section 2.1). The total tool path length including the linking segments for the LSM regions is about 2698 mm while that value for HSM regions is about 3190 mm. The fact is, for the HSM regions, as the tool path is G^1-continuous and the material removal rate is almost constant, the feed speed can be set to a large value, thus the machining efficiency can be improved.

The proposed method has also been tested on a hand pocket used by Elber, Cohen, and Drake (2005). Their path based on MATHSM is also shown in Figure 2(d). From Figure 11, it can be seen that there are 21 HSM regions and the largest HSM region is found on the palm of the hand. With our method, only a few non-cutting tool paths are generated. These non-cutting tool paths (dashed lines) are used to link the centres of the HSM regions back onto CPO path, and can be covered with

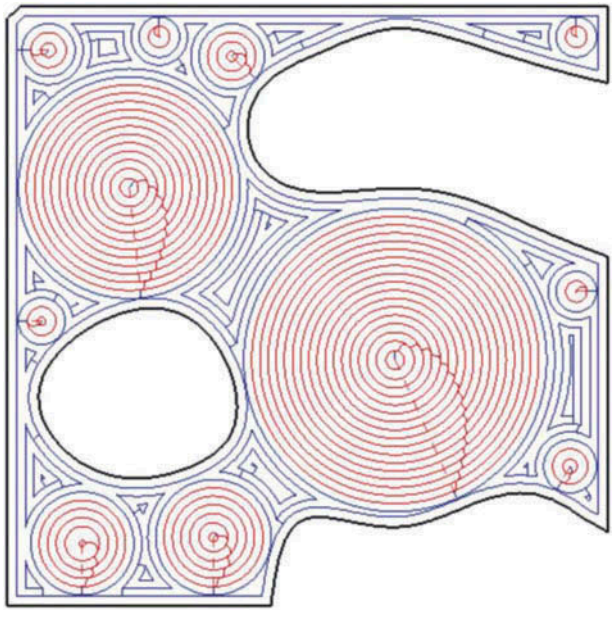


Figure 10. HSM tool path for a pocket with an island inside.

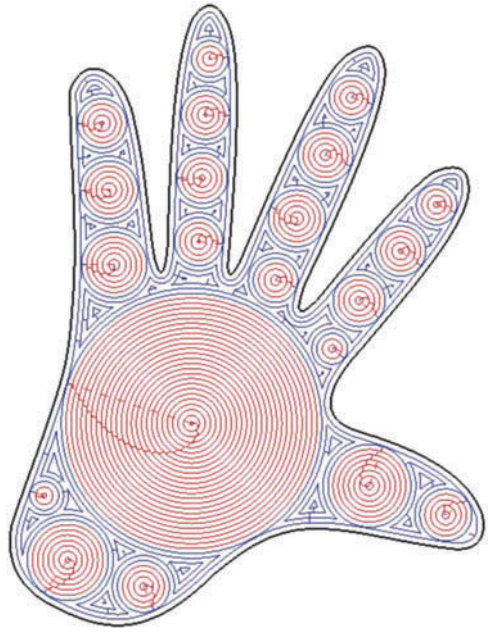


Figure 11. HSM tool path for a hand pocket.

the rapid traverse function of NCUs (such as G00 function). However, as has been discussed in Section 1.1, with the MATHSM method, about half of the generated tool path is non-cutting. For this example, as shown in Figure 11, no tool retraction is needed with our method, and the HSM regions takes about 47% of the whole pocket area.

Another example is tested on a leaf pocket as shown in Figure 12. This pocket is special because the leaf boundary is saw-like. If not optimised, sharp angles will pass onto all the descendant CPO curves, as shown in Figure 12(a). These sharp angles are nightmares to HSM and must be avoided in HSM environment. Figure 12(b) shows the HSM tool path generated by PowerMILL 10.0. From Figure 12(b), it can be seen that interior sharp angles are connected with arcs during the offset process; however, the exterior sharp angles are left untreated and inherited by the descendant CPO curves. On the CPO tool path with sharp angles, if HSM is applied, excessive chatter vibrations and tool wears might be caused. Figure 12(c) shows the HSM tool path generated by our method. In Figure 12(c), the leaf pocket is subdivided into HSM and LSM regions. In LSM regions, sharp angles are allowed to exist because low cutting speed is applied. In HSM regions, the tool paths are smooth and the material removal rate is constant, so high cutting speed is applied. As a result, with the proposed HSM tool path, the average cutting speed is high and the tool wears are controlled. For this pocket, the HSM regions take about 54% of the whole pocket area.

Table 1. Machining time statistics in VERICUT.

Method	Path length (mm)	Feed rate (mm/min)		Machining time (min:sec)
PowerMILL	9029	1000		8:23
		LSM	HSM	
proposed method	8586	1000	1000	8:08
		1000	2000	5:39
		1000	3000	4:49

Finally, the tool paths in Figure 12(b) and (c) are imported and simulated in VERICUT, as shown in Figure 13, in which Figure 13(a) shows the cutting result of the tool path by PowerMILL and Figure 13(b) shows the cutting result of the tool path by the proposed method. In Figure 13, the red path segments are circular arcs while the green ones are lines. Arcs are both used to smooth the tool path. The feed rates for the simulations in VERICUT are listed in Table 1. In Table 1, constant feed rate is set along the tool path by PowerMILL; for the tool path by the proposed method, different feed rates are set in the LSM and HSM regions. As the tool path in HSM region is very smooth, high feed rate can be set. As a result, the

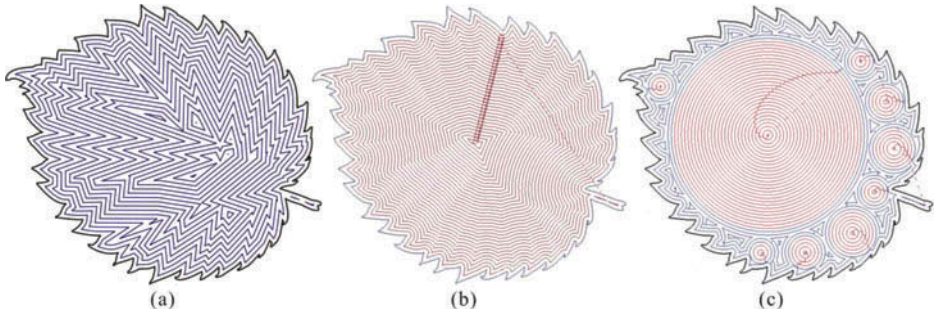


Figure 12. A leaf pocket: (a) conventional CPO curves without optimisation; (b) HSM tool path by PowerMILL; (c) HSM tool path by the proposed method.

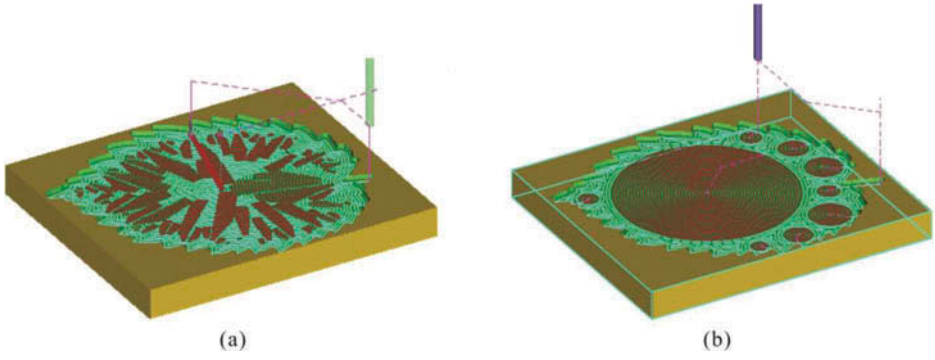


Figure 13. Cutting simulation in VERICUT: (b) PowerMILL; (c) the proposed method.

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machining efficiency can improved by using the proposed method.

5. Conclusions

HSM demands the tool path be smooth and the material removal rate be constant. Most of the existing methods attempt to achieve the first demand by sacrificing the second one. For HSM of an arbitrary pocket, the best thing is to fill the whole pocket region with a single form of continuous curves that satisfy both the above two demands. However, in reality, it is nearly impossible to find such curves. In this paper, a compromise is made by distributing not the whole but some parts of the pocket with HSM tool paths. For the rest parts, conventional CPO paths are filled. Based on the consideration that circular arcs are the simplest form of smooth curves and are acceptable by most of the NCUs, circular arcs are used for the HSM path elements. The proposed tool path generation process starts with dividing the pocket into several circular HSM regions. These circular regions are actually the MICs of the pocket and are found in an iterative offset manner. Filled in each HSM region are a series of concentric circles, the radii of which are controlled so that the cutter engagement area remains constant. These concentric circles are then smoothly linked with arcs, forming the HSM tool path. In our test examples, for an input pocket, HSM regions could take up to 50% of the whole pocket. As very high cutting speed is applied in HSM regions, the average machining efficiency can be improved.

Still, in HSM regions, the material remove rate is non-constant on the linking arcs. It is hoped that an Archimedean spiral curve that satisfies the constant material removal rate demand can be used to fill the HSM regions. But to formulate such an Archimedean spiral curve is still a challenge. In this work, the authors only focus on the smoothness of the tool path in HSM regions. As a matter of fact, existing corner rounding techniques can be used to smooth the rest CPO paths.

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